

# Ekansh Aryan Sinha

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## PROFESSIONAL SUMMARY

Quantitative Analyst fresher with B.E. Electronics and Communication Engineering and MSc Economics background, combining Python-based analytics, derivatives knowledge, risk modeling, and software implementation experience. Hands-on exposure to options, futures, Black-Scholes, Greeks, implied volatility, CAPM, VaR, ARIMA-GARCH, market-data-driven workflows, REST APIs, debugging, and automation.

## EDUCATION

**Birla Institute of Technology and Science Pilani, Hyderabad Campus** Aug 2021 - May 2026  
B.E. Electronics and Communication Engineering and MSc Economics CGPA: 8.03/10  
**Relevant Coursework:** Derivatives and Risk Management, Financial Risk Analysis and Management, Security Analysis and Portfolio Management, Business Analysis and Valuation, Econometric Methods, Applied Econometrics, Money Banking and Financial Markets, Database Management Systems, Data Structures and Algorithms

## EXPERIENCE

**MerQube** Bangalore, India  
*Quantitative Analyst Intern* Jan 2026 - Present

- Built and enhanced Python-based pricing and analytics workflows for derivatives-linked financial products.
- Worked with options, futures, implied volatility, Black-Scholes concepts, Greeks, and market-data-driven calculations to support quantitative product workflows.
- Improved debugging, validation, and automation of financial model implementation across research and production-style workflows.

**Nykaa** Bangalore, India  
*Software Development Engineer Intern* Jul 2025 - Dec 2025

- Built a FastAPI microservice for mock APIs, reducing test setup effort by 40%.
- Developed data synchronization workflows for hierarchical datasets across multiple backend modules.
- Added logging, metrics, and error-handling support to improve debugging, observability, and reliability of backend workflows.

## PROJECTS

**Quantitative Risk Modeling: CAPM, ARIMA-GARCH, and VaR Simulation** Mar 2024 - Apr 2024  
*Python, R, time-series analysis, risk modeling*

- Performed risk analysis on four companies using CAPM for beta estimation and ARIMA-GARCH models for return and volatility forecasting.
- Implemented Python-based Value-at-Risk simulation to estimate portfolio downside losses across multiple confidence levels.
- Analyzed volatility behavior, risk exposure, and portfolio downside scenarios using historical market data.

**Derivatives and Risk Management: Options Strategies and Hedging** Mar 2023 - Apr 2023  
*Options, Excel, technical indicators, risk management*

- Designed and evaluated Calendar Spread, Bear Put Spread, Iron Condor, and Butterfly Spread option strategies.
- Analyzed payoff behavior, hedging implications, and risk-return characteristics across different market scenarios.
- Used RSI, ADX, Bollinger Bands, and financial ratios to study Bajaj Finance and Syngene International across four years of data.

**Impact of Bilateral Relations on Mergers and Acquisitions** Jan 2025 - Mar 2025  
*NLP, sentiment analysis, Google News API*

- Developed a VADER sentiment-based proxy for bilateral relations to analyze news coverage around M&A transactions.
- Collected and processed six months of media data per deal using Google News API to evaluate geopolitical sentiment, investor confidence, and deal longevity.

## SKILLS

**Programming and Tools:** Python, C++, SQL, Git, Excel, Jupyter Notebook  
**Data Analysis:** NumPy, Pandas, exploratory data analysis, basic data preprocessing  
**Quantitative Finance:** Options, futures, derivatives, Black-Scholes model, Greeks, implied volatility, VaR, CAPM, portfolio theory  
**Risk and Time Series:** ARIMA-GARCH modeling, volatility analysis, portfolio risk analysis, basic backtesting  
**Software Engineering Exposure:** REST APIs, FastAPI, debugging, workflow automation, logging, error handling

## CERTIFICATIONS AND ACHIEVEMENTS

- Machine Learning Specialization by Andrew Ng; Google Cloud Fundamentals: Core Infrastructure.
- Solved 500+ LeetCode problems across data structures and algorithms.